12

Veno-Venous ECMO Physiology

Now we are at the point where we can start to apply our discussion of physiology to specific extracorporeal membrane oxygenation (ECMO) configurations. As you will see, even though they both use the same pump, circuit, membrane oxygenator, and cannulas, our two flavors of ECMO, veno-venous ECMO and veno-arterial ECMO, provide very different, support. You will get a sense of the subtleties related to these modalities in the chapters to follow.

In this chapter, we will focus on veno-venous ECMO, otherwise known as respiratory ECMO or VV ECMO.

VV ECMO: THE BASICS

Let's start this discussion by reminding ourselves of the configuration of VV ECMO. When we refer to VV ECMO, we are draining deoxygenated blood from the venous system, running it through our membrane oxygenator, and returning the oxygenated blood to the venous circulation where it gets circulated to the body through the native right ventricle (Fig. 12.1).

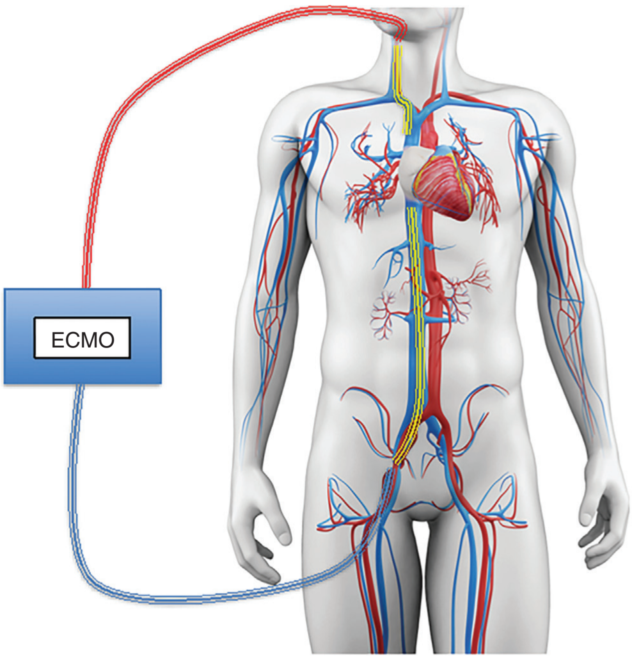

FIG. 12.1 Veno-venous ECMO. (Modified from SciePro/Shutterstock.com.)

The mechanism allows for improvement of oxygenation and ventilation/CO_2 removal parameters in patients with respiratory failure. We will further our understanding of this mechanism and physiology, exploring the rationale for the beneficial and the potential adverse hemodynamic and respiratory effects of VV ECMO.

MECHANISM OF HEMODYNAMIC/RESPIRATORY BENEFITS OF VV ECMO

Let's start by reviewing the physiologic rationale for VV ECMO. Remember that ECMO does not cure any of the inciting etiologies. Rather, it is only effective to the extent that it mitigates the toxicity related to the support that is required in response to an insult. Thus, even if the physiologic rationale is intriguing, it is only relevant in as much as it translates to an improvement in the outcome of the patient.

With that in mind, we will now explore the two primary mechanisms of benefit related to VV ECMO support, improvement of shunt and improvement of right ventricular hemodynamics.

Let's review each.

EFFECT OF VV ECMO ON SHUNT PHYSIOLOGY

As discussed in Chapter 4, while there are a variety of mechanisms for hypoxia (hypoventilation, diffusion defect, low inspired oxygen levels, intracardiac shunt), shunt physiology plays an important role. When there is worsening mismatch at the alveolar/blood vessel interface leading to V/Q of less than 1, blood shunts across the pulmonary circulation without participating in gas exchange (Fig. 12.2).

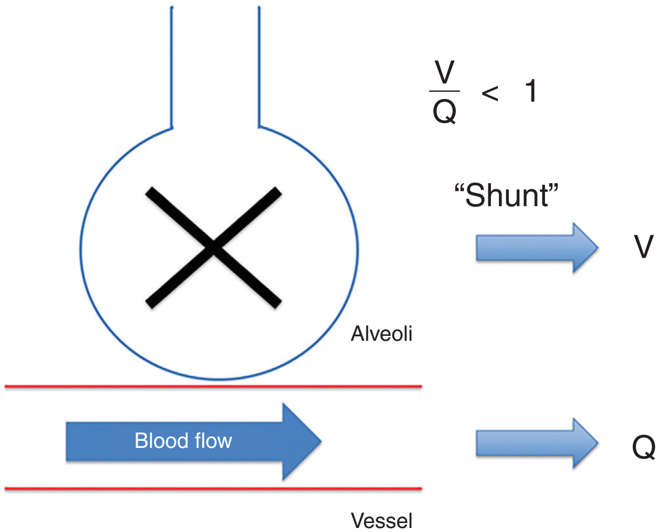

FIG. 12.2 Shunt physiology as a cause of hypoxia

This is especially important because as the percentage of shunt increases, the lungs become less responsive to administration of supplemental oxygen, and positive pressure is required to decrease the shunted alveoli.

The effect of positive pressure is relevant to our discussion with ECMO because as respiratory failure worsens, the increasing shunt and decreasing pulmonary compliance combines to require escalating ventilator pressures, and worsens the potential for damage to the lungs due to the ventilator in the form of barotrauma/volutrauma.

Shunt in VV ECMO behaves much differently.

In this case, even though blood is still being shunted across the pulmonary circulation without participating in gas exchange in the lungs, some of this blood is now already oxygenated from the ECMO circuit, and the shunted blood is oxygenated (Fig. 12.3).

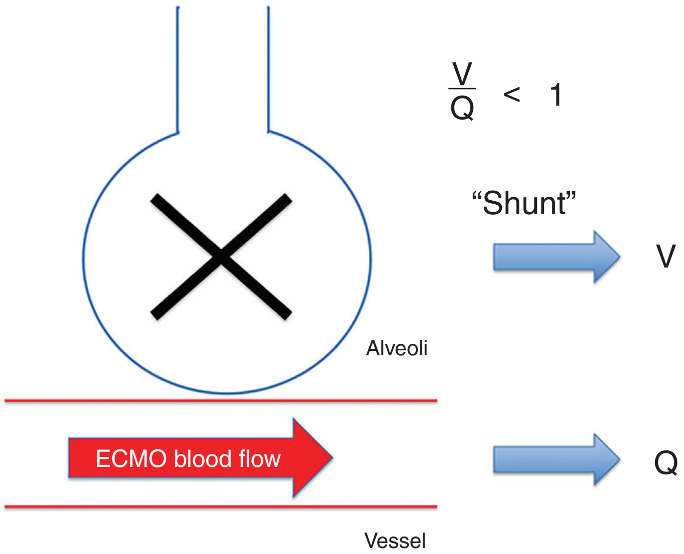

FIG. 12.3 The effect of shunt in VV ECMO

The result is that shunt fraction is improved even if V:Q matching is not. In fact, VV ECMO may even worsen V:Q matching by decreasing hypoxic vasoconstriction and increasing perfusion to damaged potions of the lungs.

This may seem like an esoteric point. You may be asking – why is this important to the patient?

Since shunt fraction is less of a priority while on ECMO, you do not need to improve shunt fraction with higher and potentially damaging airway pressures.

The extent that you can reduce elevated/damaging ventilator pressures/volumes with ECMO will determine the extent that it is able to benefit your patient.

HEMODYNAMIC EFFECTS OF VV ECMO

To better understand the hemodynamic effects of VV ECMO, let's start with the hemodynamic effects of hypoxia. If oxygen levels precipitously drop, how does this affect the cardiac output and blood pressure? To help answer, let's imagine a patient in respiratory failure who has a cardiac arrest due to a respiratory arrest. What is the typical presentation?

Usually, if you were to review the telemetry and monitor, the presentation is sinus rhythm, followed by a bradycardia, followed by hypotension, followed by arrest. Why this pattern?

The answer is that hypoxia causes hypoxic vasoconstriction as an adaptive mechanism to overcome V/Q mismatch. This vasoconstriction raises right ventricular afterload, which will drop right ventricular output leading to conduction abnormalities (bradycardia) followed by cessation of right ventricular output altogether (arrest).

You may also recall from Chapter 2 the other causes of elevated pulmonary vascular constriction that can worsen right ventricular afterload and precipitate right heart failure:

↓O_2
↓pH
↓temperature
↓pulmonary compliance
↑CO_2
↑α-adrenergic tone

Now let's return to what happens during VV ECMO – where blood is returned to the right atrium with high levels of O_2, low levels of CO_2, and is temperature controlled. The effect can be a rapid improvement in hemodynamics as the oxygenated blood reduces pulmonary vasoconstriction, decreases right ventricular afterload and eventually increases right ventricular output, leading to improved left ventricular filling, and improvement in overall cardiac output.

This hemodynamic effect of VV ECMO can lead to an improvement in DO_2 that is just as significant as the infusion of oxygenated blood via the circuit.

The degree that VV ECMO improves hemodynamics depends on the degree to which cardiac output is limited by right ventricular afterload due to hypoxia/hypercapnea. Persistent hypotension following initiation of VV ECMO should raise suspicion and prompt investigation for other causes of shock, namely vasodilation and sepsis.

It is difficult to predict the hemodynamic effect on right ventricular cardiac output that can be anticipated from the initiation of VV ECMO. At this point, it is important to be aware of the potential for improvement and to factor the anticipation of this improvement into the prediction of how ECMO will improve the patient with hypoxic respiratory failure. Understanding the physiology behind patients who respond and those who do not respond will be an essential part of selection for ECMO as well as management while on ECMO.

This is a nuanced way of thinking about VV ECMO support, but will ultimately equip you with a better sense of our sweet spot (patients who will do well with ECMO who would do poorly without ECMO) rather than just selecting patients based on specific respiratory parameters (P:F, OI, ventilator requirements, etc.).

Mechanism of Adverse Respiratory/Hemodynamic Effect on VV ECMO

When considering ECMO and the physiologic effects of ECMO we must always consider the harmful effect of ECMO support. Usually we do this by thinking about the adverse events that are often reported for ECMO – bleeding, clotting, stroke, damage to underlying structures. However, we should also consider the physiologic risks of support, particularly what limits the effectiveness of support and the degree of toxicity of conventional support that may need to be incorporated into the care as a result.

Limitations to the Respiratory Effects of VV ECMO

Said another way, what are the physiologic mechanisms behind hypoxia for patients on VV ECMO.

Wait, hypoxia on VV ECMO?

You read that right. Although VV ECMO is designed to mitigate hypoxia, hypoxia can be quite common. Let's explore why.

For a patient who is not on ECMO, as the lungs become progressively more compromised, and shunt is increasing despite all interventions, you can anticipate more blood to be shunted across the lungs, such that the saturation of blood entering the left atrium approximates the saturation of the blood leaving the right ventricle (Fig. 12.4).

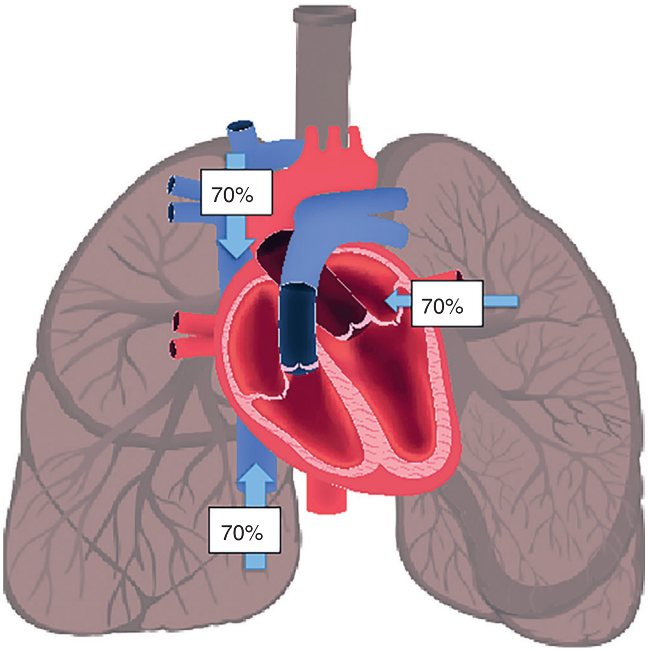

FIG. 12.4 As shunt increases, the saturation of blood returning to the left heart from the lungs approximates the SvO_2. (Modified from YanGe_Tam/Shutterstock.com.)

This will then lead to a decrease in the saturation of the venous blood returning to the heart, which in turn drops the saturation of the blood returning from the lungs, leading to a rapid deterioration.

ECMO can mitigate this downward spiral by draining the venous blood that would be going across the shunted pulmonary blood vessels and replacing it with oxygenated blood. To the extent that the drainage/return of this blood is greater than the native cardiac output, this will result in a patient oxygen saturation of 100% (Fig. 12.5).

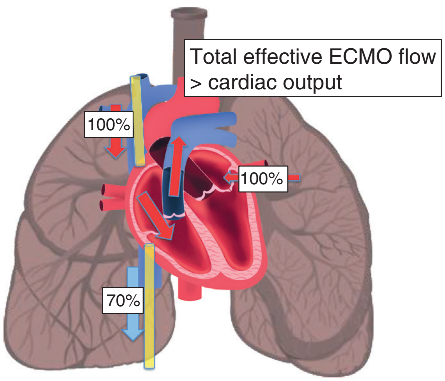

FIG. 12.5 The effect of shunt on ECMO when ECMO blood flow is higher than cardiac output. (Modified from YanGe_Tam/Shutterstock.com.)

However, let's consider this same clinical situation, except now, where the native cardiac output picks up and is greater than the ECMO blood flow by a factor of 2:1 (with a right ventricular output say of 8 L compared to a maximum effective achievable ECMO blood flow of 4 L). In this case, half of the blood that is being circulated across the shunted pulmonary circulation is 100% oxygenated while the other half is shunted venous blood that is 70% saturated.

In this case, the expected saturation of blood returning to the left heart and consequently the expected patient saturation would be 85%, the average between the blood returning from the oxygenator (100% saturated) and the blood that is not flowing through the oxygenator (70% saturated) (Fig. 12.6).

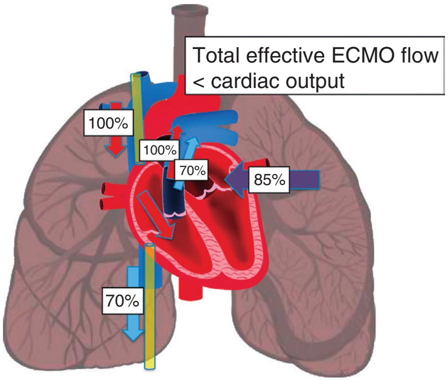

FIG. 12.6 The effect of shunt on ECMO when cardiac output is higher than ECMO blood flow. (Modified from YanGe_Tam/Shutterstock.com.)

Continuing this thought experiment, as the ratio of cardiac output to achievable ECMO blood flow continues to increase, the percentage of deoxygenated blood returning to the left heart increases, such that the patient saturation starts to approximate the saturation of the venous blood (70% in this case).

OTHER VARIABLES CONTRIBUTING TO OXYGENATION IN VV ECMO

What this example is describing is a relationship between two competing circulations: the native circulation/lungs and the ECMO circulation/membrane oxygenator. The greater the contribution of the native circulation, the more the saturation of the patient will approximate the saturation of what the lungs can contribute outside of ECMO (SaO_2 native lungs). If we were to write this relationship as an equation, it might look something like this:

$$\begin{aligned} SaO_{2\,patient} &= SaO_{2\ oxygenator}\left[\frac{\text{Effective ECMO flow}}{\text{Cardiac output}}\right] \\ &+ SaO_{2\ native\ lungs}\left[\frac{\text{Cardiac output} - \text{Effective ECMO flow}}{\text{Cardiac output}}\right] \end{aligned}$$

This is the central equation for understanding hypoxia on VV ECMO. The more you can understand the concepts behind this equation, the better you will have a system to troubleshoot hypoxia. Let's evaluate each component.

SaO_2 of the Oxygenator

This is the saturation of the blood returning from the oxygenator. Assuming a working membrane oxygenator, this can be assumed to be 100%.

Effective ECMO Flow

$$\text{Effective ECMO flow} = \text{ECMO blood flow} - \text{recirculation}$$

Remember we say effective ECMO flow, not blood flow. Effective ECMO flow is the blood flow that is actually reaching the patient. It combines the limits of the circuit/pump that were described in Chapter 10 as well as recirculation. Recirculation decreases the effective ECMO flow according to the following equation:

The challenging thing about recirculation is that it can't be quantified, only inferred/estimated based on blood gases. Recirculation should be a consideration anytime there is less than a 20% difference between the pre-oxygenator blood gas and the patient arterial blood gas.

Cardiac Output

As cardiac output (specifically right ventricular output) increases, the proportion of blood going across the lungs increases. The higher the native cardiac output compared to the effective ECMO blood flow, the closer the patient saturation will be to the native lung SaO_2.

Does dropping the cardiac output make sense? It is possible that lowering cardiac output (say with beta blockade) will result in an increased SaO_2. However, this higher SaO_2 may not contribute to a higher DO_2, since you had to trade off cardiac output to get it.

When does decreasing cardiac output make sense? When the elevated heart rate/stroke volume is not a function of the body trying to augment DO_2 – fever, pain, agitation, hypermetabolic state, excess pressors, etc. In these cases, bringing down heart rate/cardiac output may have the effect of treating the underlying cause and improving saturations.

SaO_2 Native Lungs

This is the saturation that would be returning to the left atrium if the ECMO circuit was not running. It is comprised of two variables:

1. Contribution of the lungs
2. Native venous oxygen saturation: function of DO_2:VO_2

Application of This Equation

This equation demonstrates the essential points but doesn't require actual calculations at the bedside. Rather than trying to calculate numbers, use this equation to structure your approach and infer what is going on with the patient. Let's use an example.

Say you are taking care of the following patient:

44 year old man on day 3 of VV ECMO (cannulated by femoral-internal jugular vein approach)
Blood flow 3.5 L, 1800 RPM, sweep gas flow 3 L, FiO_2 100%
Ventilator: AC/VC, 70% FiO_2, TV 300 mL, PEEP 10, rate 20
ABG: 7.39, 41, PaO_2 45, **SaO_2 75%**

If we want to improve this saturation, there are only a few options.

1. **Increase effective ECMO blood flow**. Limitations of this strategy include limits of the pump, oxygenator, resistance of the circuit, effects of RPM (hemolysis), and recirculation.

If blood flow is increased, attention should be paid to excessive circuit pressures, evidence of hemolysis (LDH, hemolysis, dark urine), and evidence of recirculation (elevation of pre-oxygenator SvO_2).

2. **Decrease cardiac output**. Can consider but this may not help DO_2. Attention should be paid to evidence of worsening DO_2:VO_2 ratio (such as decreased SvO_2, lactic acidosis, or decreased urine output).
3. **Increase SaO_2 of native lungs**. This can be done through augmentation of the mechanical ventilation settings or augmentation of native DO_2:VO_2. Higher ventilator settings may risk toxicity of the ventilator. Since DO_2 is function of SaO_2, cardiac output, and hemoglobin, this comes in the form of administration of blood.

Now you are in a position to compare the competing circulations (ECMO v. native) and to evaluate the effect of your interventions. Consider the following stepwise approach to the patient with hypoxia on VV ECMO in Fig. 12.7.

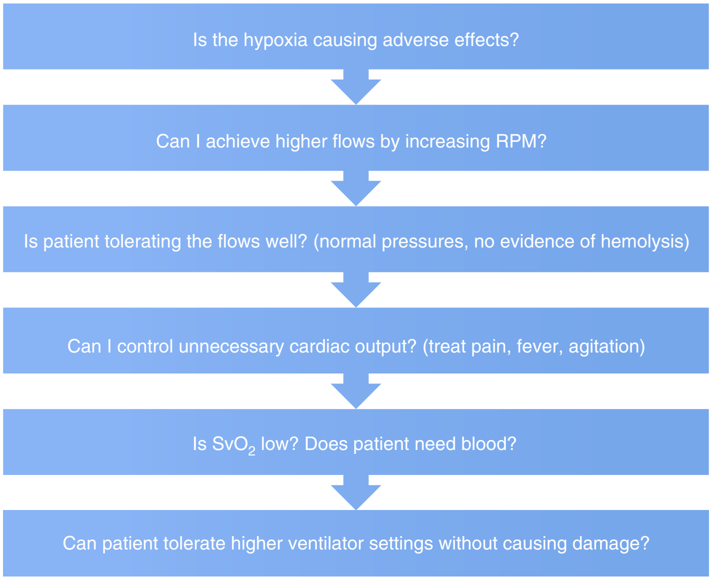

FIG. 12.7 Stepwise approach to the patient with hypoxia on VV ECMO. *RPM*, Revolutions per minute.

You can see how you are now adopting a much more nuanced approach than only increasing the ventilator or worse, doing everything at once with no ability to evaluate the effect and understand the physiology of what is going on with the patient.

LIMITATIONS TO THE CARDIAC/HEMODYNAMIC EFFECTS OF VV ECMO

Recall the hemodynamic rationale for VV ECMO is improvement of right ventricular afterload, through decrease in hypoxia/hypercapneia-mediated vasoconstriction. However, right ventricular failure can be common in VV ECMO, even with optimization of O_2/CO_2 levels.

The likely etiology of right ventricular failure/elevated pulmonary pressures in VV ECMO is vasoconstriction due to damaged lung that is mediated by cell-mediated/inflammatory mechanisms and poor pulmonary compliance. It is possible that this allows for improvement of V/Q matching in the presence of chronic hypoxia. In these cases, simply correcting O_2/CO_2 levels of the blood circulating to the pulmonary circulation does not improve the right ventricular afterload or function.

Rather, VV ECMO flow may serve to make this right ventricular function worse. In chronic pulmonary hypertension due to lung disease, elevated right ventricular afterload due to chronic hypoxia decreases overall cardiac output leading to an adaptive drop in filling pressures. By contrast, in a patient on VV ECMO, when the right ventricle fails due to worsening pulmonary disease/inflammatory mediators, there is no drop in filling pressures, as the flow of blood returning from the ECMO circuit remains constant.

The effect of a failing right ventricle on VV ECMO is less effective ECMO function. This happens in the following ways:

1. Worsening right ventricular output, with less ECMO blood flow circulating to the systemic circulation
2. Higher right ventricular filling pressures, with eventual worsening of right ventricular systolic function as well as left ventricular systolic function due to ventricular interdependence (see Chapter 2)
3. Less blood that is able to go forward past the right ventricle, leading to increasing recirculation and decreased effective ECMO blood flow

PUTTING IT TOGETHER: THE RISK-BENEFIT OF VV ECMO SUPPORT

We have spent a lot of time in this chapter discovering the rationale and limitations of VV ECMO support. This is not just a theoretical exercise, but rather a description of the physiologic subtypes of patients who do well with VV ECMO and patients in whom the benefits of VV ECMO are limited.

In all cases, there is a risk-benefit of ECMO support that must be considered. There are limitations to ECMO support as well as adverse effects. Just as we have explored the dose-related response of conventional care when it comes to vasopressors, fluids, and ventilation support, there is a dose-related response to ECMO support. This exists in terms of blood flow, duration of support, and hemodynamic effects.

We can conceptualize the overall risk profile of ECMO support as illustrated in Fig. 12.8, imagining a clinical improvement that exists to a point, after which, harmful effects of increasing blood flow take over, and there is a point of diminishing returns.

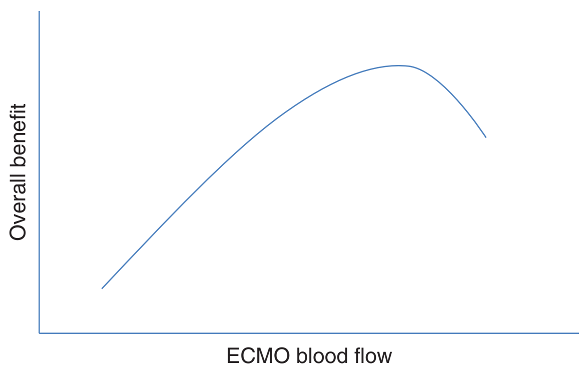

FIG. 12.8 ECMO blood flow risk-benefit

The decision about whether ECMO support should be titrated up or down needs to be put into the context of the benefit and harm, and can be elucidated by attention to the response to any changes.

SUGGESTED READING

Darryl, A., Bacchetta, M., & Brodie., D. (2015). Recirculation in venovenous extracorporeal membrane oxygenation. *ASAIO Journal*, *61*(2), 115–121.

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Patel, B., Arcaro, M., & Chatterjee, S. (2019). Bedside troubleshooting during venovenous extracorporeal membrane oxygenation (ECMO). *Journal of Thoracic Disease*, *11*(Suppl 14), S1698.